Geographic Disparities in Alzheimer's Disease and Related Dementia Mortality in the US: Comparing Impacts of Place of Birth and Place of Residence

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Abstract

Objective: Building on the hypothesis that early-life exposures might influence the onset of Alzheimer's Disease and Related Dementia (ADRD), this study delves into geographic variations in ADRD mortality in the US. By considering both state of residence and state of birth, we aim to discern the comparative significance of these geospatial factors.
Methods: We conducted a secondary data analysis of the National Longitudinal Mortality Study (NLMS), that has 3.5 million records from 1973-2011 and over 0.5 million deaths. We focused on individuals born in or before 1930, tracked in NLMS cohorts from 1979-2000. Employing multi-level logistic regression, with individuals nested within states of residence and/or states of birth, we assessed the role of geographical factors in ADRD mortality variation.
Results: We found that both state of birth and state of residence account for a modest portion of ADRD mortality variation. Specifically, state of residence explains 1.19% of the total variation in ADRD mortality, whereas state of birth explains only 0.6%. When combined, both state of residence and state of birth account for only 1.05% of the variation, suggesting state of residence could matter more in ADRD mortality outcomes.
Conclusion: Findings of this study suggest that state of residence explains more variation in ADRD mortality than state of birth. These results indicate that factors in later life may present more impactful intervention points for curbing ADRD mortality. While early-life environmental exposures remain relevant, their role as primary determinants of ADRD in later life appears to be less pronounced in this study.

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Introduction

Alzheimer's Disease and Related Dementia (ADRD) significantly impacts numerous American families. Approximately 5 million Americans over age 65 have Alzheimer's Disease – a subset of those with ADRD – and these numbers are expected to swell to nearly 14 million by 2050 (Hebert et al., 2013). Roughly one in five American women or one in ten American men are anticipated to face dementia in their lifetimes (Chêne et al., 2015). Fifteen percent of Americans over the age of 70 have dementia, with an estimated financial burden of just under $30,000 annually per person, excluding the expenses related to informal caregiving (Hurd et al., 2013). Individuals shouldering an average of over $6,000 in out-of-pocket expenses (Hurd et al., 2013). The substantial economic and health burdens underscore the urgency to understand the fundamental causes of ADRD, one of the primary causes of mortality in the US.

Numerous studies have identified specific individual risk factors that contribute to the risk of ADRD mortality. Individual characteristics including lower educational attainment (Korhonen et al., 2020) , poorer health (e.g., high blood pressure, physical inactivity, obesity, diabetes, etc.) (Chen et al., 2023; Omura et al., 2022), rural residence (Cross & Warraich, 2021), and genetic predispositions (Bellenguez et al., 2022) are associated with elevated risks of ADRD mortality. However, some of these individual characteristics could systematically vary by place, suggesting local environments could play a role in ADRD mortality. For example, although individuals choose their education attainment, compulsory schooling laws in the late 19th century and early 20th century in the US varied by state and meaningfully altered individual attainment (Clay et al., 2012, 2021; Lleras-Muney, 2002; Stephens & Yang, 2014). Moreover, health behaviors and healthcare systems are subject to significant geographic disparities (Tsui et al.,

2020), which indicates that regional variations in healthcare access and health behaviors may contribute to an increased risk of ADRD mortality.

In fact, the local environment itself likely also matters for dementia risk, especially Alzheimer's disease and mortality. Case and Paxson (2009) found that lower local infant mortality rates around age two in the US, indicative of the local health environment, were linked with improved cognition at older ages. Similarly, Topping et al. (2023) showed that early life state-level infant mortality rates were associated with ADRD mortality, especially among those under 65 years of age. On the policy front, Medicaid, primarily a state-administered health insurance program, has been linked to several health outcomes. Miller et al. (2019) found recent ACA-related expansions reduced mortality. Earlier expansions also appear to improve population health (Boudreaux et al., 2016; Miller & Wherry, 2019). Access to Food Stamps or the Supplemental Nutrition Assistance Program also improved health years later and reduced mortality rates (Heflin et al., 2019; Hoynes et al., 2016; Jones, 2020). While improved health could decrease dementia prevalence, lower mortality could increase prevalence by eliminating the competing risk. In any case, policies at the state or local levels can theoretically influence both the risk and prevalence of dementia.

The relative importance of early-, mid-, or late-life environments for ADRD mortality risk remains uncertain. Supporting the case for the importance of early-life environments, lower educational attainment is one of the most widely recognized risk factors for dementia (Cook & Fletcher, 2015; Harrison et al., 2015; Sattler et al., 2012; Wilson et al., 2009), as well as ADRD mortality (Garcia et al., 2021; Korhonen et al., 2020). It is hypothesized that higher education is linked with lower dementia prevalence and its related mortality via increases in cognitive reserve (Stern, 2009). Importantly, primary and secondary education is largely determined in early-life,

and historic state policies in the late 19th and early 20th altered attainment in cohorts currently at risk for dementia (Clay et al., 2012, 2021; Lleras-Muney, 2002; Stephens & Yang, 2014). More directly, a few studies found that while both residing in and being born in the US "stroke belt" states is a strong predictor of dementia, birth location may dominate (Glymour et al. 2011; Topping et al. 2021a).

A growing body of research also underscores the role of early-life environments in shaping health outcomes. Cognition is related to the local disease environment when an individual was two years old (Case & Paxson, 2009). Exposure to food stamps in early childhood (i.e., ages 0 – 5) decreased metabolic syndrome (Hoynes et al., 2016), while early-life exposure to Medicaid around these same ages improved adult health (Boudreaux et al., 2016) and decreased mortality (Goodman-Bacon, 2021). In the US, however, most major social safety-net expansions started in the 1960s. This suggests that many individuals impacted by these policies during childhood might be too young to observe dementia onset and mortality. One exception is the Mother's Pension Program—a cash transfer program for mothers between 1911 and 1935 that varied state-by-state—that meaningfully altered longevity, educational attainment, and income for beneficiaries' sons (Aizer et al., 2016). In the early 20th century, poverty was widespread in the US, exacerbated by the Great Depression. However, the impact of the Great Depression varied across the US, with the South Atlantic states enduring a milder downturn and the Mountain states suffering more, influenced by industrial composition and long-term regional trends in employment (Margo, 1993; Rosenbloom & Sundstrom, 1999).

In addition to early exposure, individuals are likely to be exposed to different environments during middle and old age. Approximately one-third of people in the US reside in a different state than where they were born (Kahn & Pearlin, 2006). Therefore, to better

understand the contribution of early-life environments to ADRD mortality risk, it is required to consider the role of later-life environments simultaneously (Fletcher et al., 2022). For example, Miller et al. (2019) found that ACA Medicaid expansions reduced mortality among individuals aged 55-64. Expansions of the Earned Income Tax Credit (EITC) in the 1990s increased employment and mental health among mothers (Evans & Garthwaite, 2014; Hoynes & Patel, 2018) and improved the overall health status (Lenhart, 2019). The fact that (i) many US safety-net programs started after 1960, (ii) exposure to the social safety-net in later-life offers material and health advantages, and (iii) data limitations on early-life exposure to dementia risks, all curb researchers' ability to detect early-life exposure effects. This would lend support to the notion that later-life environments may matter more for determining ADRD mortality, particularly among individuals currently at risk of dementia. Moreover, regional variation in competing causes of death (deaths from other causes such as infection or cardiovascular disease) during the 1970s-1990s, potentially attributable to varying levels of regional public investment (Mensah et al., 2017), may reflect in regional variation in ADRD mortality later in life.

Our study investigates the relative importance of early- versus late-life environment contexts in explaining ADRD mortality risk. This research holds critical implications for policymakers endeavoring to discern the optimal life course stages for policy interventions and for researchers or practitioners seeking insights into how ADRD mortality risk progresses over the lifespan. Specifically, our study considers whether birthplace and/or place of residence are most important when modeling ADRD mortality.

Although there are a few studies on geographic variation in mortality, evidence on the relative importance of state of residence versus state of birth is mixed (Finkelstein et al., 2021; W. Xu et al., 2020). Xu et al. (2021) found that the state of birth accounts for almost twice the

variation in cardiovascular disease mortality compared to the state of residence. Conversely, another study revealed that state of birth and state of residence contributed the similar amount of variation in all-cause mortality (Xu et al., 2020). With respect to Alzheimer's disease mortality specifically, one study documented the association between local-level infant mortality rates in early life and death from ADRD in the US (Topping et al., 2023). However, Topping et al. (2021b) reported that an individual's state of birth explains relatively little variation in Alzheimer's disease mortality—providing evidence against early-life factors affecting Alzheimer's disease mortality. The findings suggested that almost all the variation in Alzheimer's disease mortality related to the state of birth (about 4%) is accounted for by the state of residence in later life. Our study offers several key improvements. First, we broaden the definition of dementia to include all dementia types, not just Alzheimer's disease. Second, we use a more representative sample from across the United States. Finally, we have approximately four times as many ADRD deaths recorded. With these improvements, our results offer continued support for the notion that later-life contexts matter more than early-life contexts in ADRD risk for current cohorts at risk of dementia.

Data and Methods

Data

We use data from the restricted version of the National Longitudinal Mortality Study (NLMS). The NLMS has around 3.5 million records from the 1973-2011 Current Population Survey (CPS) and 1980 Census. These cross-sectional surveys (i.e., CPS or Census) are linked to over 0.5 million deaths. The cross-sectional surveys are referred to as NLMS cohorts (NLMS, 2014). Across all NLMS cohorts, approximately 5% of available respondents have a "fail-edit" with mortality data, meaning it is not feasible to verify their vital status, and they are not part of the

NLMS (Table 2, NLMS, 2014). The key variables we use from the NLMS to complete our analysis are the individual's age and state-of-residence when completing the survey (i.e., CPS or Census), place of birth, race/ethnicity, sex, educational attainment, birth cohort, and cause of death (if applicable).

Our sample includes individuals born in or before 1930 and observed in NLMS cohorts from 1979 – 2000. The 1973 NLMS cohort is excluded because it requires 6 years of follow-up prior to the start of the National Death Index in 1979 (NLMS, 2014). All individuals are followed until death or the end of 2002, whichever is earlier. These restrictions are necessary because place of birth at the state level is not included in NLMS cohorts from 2003 onward (NLMS, 2014) and individuals dying after 2002 rarely have place of birth recorded at the state level. Individuals included in the sample are also observed between ages 52 – 93 at the time of survey (inclusive) to capture ages most at risk of ADRD mortality[1]. Finally, included individuals are born in, and reside in, a state with at least ten ADRD sample deaths.

The analytic sample consists of approximately 253,000 individuals. Due to the sampling structure (i.e., age at interview and using waves before 2001), over 70% of the sample dies during the follow-up period. Although dementia is a leading cause of death in the United States (J. Xu et al., 2022), this single cause of death represents just under 2% of the analytic sample (or around 2.4% of observed deaths).

[1] Most of the sample is interviewed in 1985 or earlier (and followed until death or the end of 2002, whichever is earlier). Additionally, all sample members are born in 1930 or earlier. Thus, individuals who are observed as young as 52 are, by construction, all interviewed in 1985 or earlier.

Measures

The two key explanatory variables in our analysis are state of birth and state of residence. The NLMS "Place of Birth--Final" variable characterizes individual's state of birth. It consists of survey responses for some NLMS cohorts supplemented with administrative record linkages such as the Social Security Administration's NUMIDENT file. State of residence is the recorded Federal Information Processing Standards (FIPS) state code in the NLMS data. If early-life factors dominate, then birth location should capture more variation than current residency (and vice versa). The dependent variable is a binary indicator for an ADRD death. Among individuals identified as deceased in NLMS, cause of death is identified by ICD-9 and ICD-10 codes. Table 1 highlights the specific codes we use to classify a death as ADRD mortality. Race/ethnicity is a categorical control variable consisting of non-Hispanic White (reference), non-Hispanic Black, Hispanic, and non-Hispanic other race or ethnicity. It is constructed using NLMS data on race and Hispanic ethnicity. Sex is a binary indicator for female. Educational attainment is grouped as less than high school, high school, some college, and college graduates (reference) using the education recode in NLMS.

(Table 1 about here)

Analytic strategy

We compare the relative importance of state-of-birth versus state-of-residence using a series of random effects models (xtlogit, re in Stata). Standard covariates including sex, race/ethnicity, and educational attainment (X_i) are regressed on a binary indicator for ADRD mortality ($ADRD_i$) with an ADRD death coded as 1 and survivors or non-ADRD deaths coded as zero. Interview age (γ_i), and NLMS cohort (φ_i) fixed effects are also included as in equation (1).

(1) $$ADRD_i = \alpha_i + \beta X_i + \gamma_i + \varphi_i + \varepsilon$$

α_i is the random effect. Although coefficients for each state are not directly measured in the modeling strategy, we compare the amount of variation explained by state of birth, state of residence, or a combination of both state of birth and residence for each model using $\ln(\sigma_u{}^2)$ – a variance component for the random effects – and ρ – the proportion of total variance attributable to variance in the random effects. We additionally use the AIC and BIC to assess model quality.

Results

Our sample descriptive statistics (Table 2) demonstrate that ADRD mortality is primarily among the oldest Americans. Women and individuals from advantaged backgrounds appear more likely to survive to an age when ADRD can be the main underlying cause of death. Survivors are the youngest sample members—on average being interviewed at age 65 and last observed at age 78—while the deceased are significantly older. Individuals dying from non-ADRD causes on average are interviewed at age 71 and die 8 years later. However, people dying of ADRD are significantly older. On average they are interviewed at age 74 and die at age 85.

(Table 2 about here)

Nearly two-thirds of survivors are women, while closer to half of the deceased from non-ADRD causes are women. ADRD deaths though are predominantly among women. ADRD deaths are also primarily among non-Hispanic White individuals. While 88% of survivors are non-Hispanic White, 92% of ADRD deaths are among non-Hispanic Whites. Non-Hispanic Black individuals make up a slightly larger share of the deceased from non-ADRD causes relative to their share of survivors, but they make up a significantly smaller share of ADRD deaths relative to their share of survivors. The distribution of educational attainment among the deceased skews toward lower education levels. For example, half of the deceased from non-

ADRD causes have less than a high school degree while just 28% of survivors have less than a high school degree. People dying of ADRD however, appear slightly more educated than those deceased from other causes. Just 44% of ADRD deaths are among people with less than a high school education.

We begin with observing spatial variation in ADRD mortality across the United States in Figure 1. Similar to Akushevich et al. (2021), our figure highlights significant spatial variation in dementia mortality. The northwest, and to a lesser degree southeast, appears to have higher rates of ADRD mortality while the midsection tends to have lighter shades of blue representing lower ADRD mortality rates. Despite these broad similarities, the side-by-side maps of spatial ADRD mortality rates by state of birth and state of residence are clearly not the same. ADRD mortality is higher among individuals born in Idaho or New York relative to individuals who live in in those states. Conversely, ADRD mortality is lower among individuals born in Washington or Louisiana relative to those who live in those states.

(Figure 1 about here)

Regression results (Table 3) suggest that state of residence explains relatively more variation in ADRD mortality than either state of birth alone or a combination of state of birth and state of residence. Column 1 contains baseline estimation excluding random effects. Column 2 includes state of residence random effects while column 3 includes state of birth random effects. Column 4 includes a combination of state of residence and state of birth random effects. Both the AIC and BIC are minimized when including only state of residence random effects (column 2) while $\sigma_u{}^2$ and ρ are maximized in this specification. Table 3 shows that state of residence explains 1.19% of the total variation in ADRD mortality (column 2), whereas state of birth

explains only 0.6% (column 3). Column 4 suggests that both state of residence and state of birth explain only 1.05% of the total variation in ADRD mortality.

(Table 3 about here)

To assess the robustness of our findings, we conducted a series of sensitivity analyses. First, we examined whether our results are sensitive to the inclusion of age as a quadratic continuous variable. In Table S1 of the online supplementary file, substituting a quadratic continuous age covariate confirms the robustness of our main conclusions. Second, to address the potential information misclassification bias caused by including non-deceased participants—where participants who may develop dementia before death are classified as non-cases—we estimated models that exclude survivors from the sample. The results, presented in Table S2 of the online supplementary file, remain substantially similar to the original findings. We have also used alternative specifications that use Stata's "melogit" command that nest state of residence and state of birth random effects instead. In this case, the variance explained by state of residence is still the dominating feature.

Discussion

ADRD is a pressing public health concern, the impacts of which reverberate through individual lives, families, and the broader socio-economic fabric of the US. Investigating geographic variation in ADRD mortality is one avenue to contemplate the relative importance of early- versus late-life factors influencing later-life cognition by examining the relative importance of one's birth location versus location of residence. If early-life factors are more important, then we can expect birth location to matter more whereas if later-life factors dominate, residency could be more important. Our study results support the notion that later-life environments could matter more than early-life environments for the likelihood of dying from ADRD. Leveraging a large

sample of ADRD deaths available in restricted-use NLMS data, we find that state of birth consistently explains less variation in ADRD deaths relative to observed state-of-residence at ages 52 – 93. These results corroborate an earlier study using a smaller sample and more narrow definition of dementia mortality (Topping et al., 2021b); however, we doubt this is to be a definitive or final verdict on the question. While our results are interesting, informative, and a necessary first step to understanding the relative importance of environment contexts during the life course, future research with younger birth cohorts aging into ADRD mortality could be better equipped to address optimal life course stages for policy intervention.

These findings echo and expand upon the current discourse in the field. While early-life factors such as educational attainment, influenced by state-level policies and local environments, undoubtedly play a role in cognitive outcomes in later life, our findings suggest that where an individual spends their later years might be just as, if not more, influential. Delving deeper into potential mechanisms, several plausible explanations emerge. One of the foremost considerations is the exposure to state-specific health policies and infrastructures (Osypuk et al., 2014). Every state has its own health mandates (e.g., Affordable Care Act/Medicaid), insurance structures, and established guidelines for treating older adults. These policies, whether they pertain to preventive care, early detection, or disease management, can significantly shape the health trajectories of its older residents. Furthermore, states that have robust healthcare infrastructures, with easy access to geriatric care specialists, neurologists, and mental health professionals, might offer residents a better chance at early detection and management of ADRD (Bradford et al., 2009).

That said, it is essential to approach these findings with caution. While we have made strides in our understanding, the magnitudes of explained variation by state of residence or birth are modest. The substantial remainder of unexplained variation suggests that there are other

significant factors, beyond state of residence or birth, that influence ADRD outcomes. There are undoubtedly many risk factors that remain unknown and several known risk factors lack a causal link to dementia. These might include individual genetic predispositions, lifestyle factors, or other unmeasured environmental exposures. An asset of our study design is that we can remain agnostic on what specific early- and late-life factors contribute to our observed patterns. It implicitly incorporates both known and unknown environmental risk factors into determining whether early- or late-life matters more for ADRD mortality. In this way, the study can retain applicability even after new discoveries are made. Thus, our study serves as a foundational step, opening doors for more nuanced and multi-dimensional research into the geographical distribution of ADRD.

This study has several limitations. First, we are examining ADRD mortality, not the presence of cognitive impairment, decline, or even the presence of dementia leading up to death. To be classified as an individual with ADRD mortality, this condition must have been listed as the underlying cause of death on the death certificate. It is unlikely that all individuals with underlying dementia indications or pathology at the end of life will be listed with dementia as an underlying cause of death (Todd et al., 2013). Thus, although the populations of individuals with dementia and those dying from dementia overlap, they are distinct. NLMS is not equipped to comment on the presence of cognitive impairment or pathology, only the underlying cause of death as recorded. It is likely that some individuals with respiratory or circulatory diseases listed as the underlying cause of death may have had ADRD (Brunnström & Englund, 2009). When considering the link between pollution and dementia, a review found higher exposure is almost always linked with dementia, but the link with cognitive decline is more mixed (Peters et al., 2019). In their analysis of geographic disparities of Alzheimer's Disease mortality, Akushevich

et al. (2021) conclude that differences in filling out death certificates are an unlikely cause of geographic disparities. Their conclusion suggests this limitation may not meaningfully contribute to our results.

Second, mortality selection is an important factor. Bishop, Ketcham, and Kuminoff (2022) suggest that individuals who died would have been more likely to develop ADRD (as measured by Medicare claims data while alive) had they lived longer. We see evidence of this pattern in our own sample (Table 2). Individuals dying during follow-up without ADRD are typically younger. There's also a larger portion with lower educational attainment, and individuals who are non-Hispanic Black among those dying of non-ADRD causes relative to those dying of ADRD. Thus, ADRD mortality is concentrated among the oldest old: those who have survived long enough to acquire and die from ADRD. At the same time, one of the most consistent risk factors for cognitive impairment or developing ADRD is lower educational attainment (Cook & Fletcher, 2015; Harrison et al., 2015; Sattler et al., 2012; Wilson et al., 2009). However, lower socioeconomic status, and education in particular, is importantly also a significant risk factor for higher mortality rates (Chetty et al., 2016; Montez et al., 2019) and reiterates the fact that we are studying the presence of ADRD mortality, not cognitive impairment, decline, or ADRD indications near the end of life.

Third, we limit our sample to individuals born in 1930 or earlier. While this ensures ample time to observe mortality, no one in our sample was exposed to the social programs forming today's safety-net during their young childhood. Instead, individuals from these cohorts primarily encountered the safety-net in adulthood. Two potential exceptions to this generalization are the Mothers' Pension program and the Aid to Families with Dependent Children (AFDC) program, which was introduced in 1935. In the case of the AFDC, the

youngest in our sample might have been exposed during childhood, but never before reaching five years of age. Later safety-net programs like Food Stamps, the predecessor of SNAP, began as a pilot in 1961 (Hoynes et al., 2016), by which time our youngest cohort was already in their 30s. Medicaid started shortly after, and the Earned Income Tax Credit did not begin until 1975. Many programs comprising the US social safety-net differ from state to state. To the degree that improved health and wellbeing from these programs is linked with ADRD mortality, in our sample it would likely only be detected by adulthood state of residence. As individuals exposed to the social safety-net as children age, future research may need to revisit this important question.

In conclusion, our study sheds light on the often-overlooked geographical nuances in the understanding of ADRD. While early-life factors and state of birth may be influential, our findings emphasize the importance of considering where one resides in their later years, at least for current cohorts at risk of ADRD. It is crucial to consider the broader implications of these findings. If the state of residence in later life is indeed important, then interventions targeted at improving the quality of life for older adults, enhancing healthcare access, or reducing environmental risk factors in states with higher ADRD prevalence could be strategic starting points. Additionally, understanding the influence of state-specific policies or practices could guide more targeted, state-level interventions, potentially leading to reduced ADRD incidence or improved care outcomes. As an effort to address ADRD progress, such insights can guide more effective, targeted strategies, bringing us one step closer to a world with a deeper understanding and, hopefully, better management of this debilitating condition.

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Table

Table 1. ICD-9 and ICD-10 codes used to identify ADRD

	ICD-9	ICD-10
Alzheimer's Disease	331.0	F00 & G30†
Vascular Dementias	290.4	F01†
Disease Dementias/Other conditions	294.1	
Unspecified Dementias	331.1-331.2	F03
Senile Dementia	290.0 – 290.3	
Other cerebral degeneration	331.8	
Frontotemporal dementia		G31.0
Senile		R54

† F01 includes F01.1, F01.3, F01.8, and F01.9; G30 includes G30.0, G30.1, G30.8, and G30.9. ICD = International Classification of Diseases; ADRD = Alzheimer's Disease and Related Dementias.

Table 2. Sample descriptive statistics

	Survivors	Non-ADRD mortality	ADRD mortality	Statistical difference between non-ADRD and ADRD mortality
Age at interview	64.52	71.09***	74.03***	***
	(.0338)	(.0276)	(.159)	
Age at end of follow-up	78.4	79.31***	85.14***	***
	(.02457)	(.02648)	(.136)	
% Female	.6429	.5124***	.657	***
	(.0022)	(.0015)	(.009)	
% non-Hispanic White	.8837	.8807+	.92***	***
	(.0015)	(.001)	(.006)	
% non-Hispanic Black	.0808	.0934***	.063***	***
	(.0013)	(.001)	(.005)	
% Hispanic	.0236	.0175***	.011***	***
	(.0006)	(.0003)	(.002)	
% Other race/ethnicity	.0119	.0084***	.006***	+
	(.0005)	(.0003)	(.001)	
% < High school	.2844	.4986***	.443***	***
	(.002)	(.0015)	(.009)	
% High school	.3996	.2999***	.332***	***
	(.0022)	(.0014)	(.009)	
% Some college	.1536	.1054***	.116***	
	(.0017)	(.0009)	(.006)	
% College	.1624	.096***	.11***	*
	(.0017)	(.0009)	(.006)	
N	73,500	175,000	4,300	

Source: Authors' calculations using NLMS data at a Federal Statistical Research Data Center under FSRDC Project Number 2338 (CBDRB-FY23-P2338-R10359).

Notes: ADRD death classifications are in Table 1. Standard errors in parenthesis. ADRD = Alzheimer's Disease and Related Dementias. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$ relative to the reference category (survivors).

Table 3. Random effects models predicting ADRD mortality

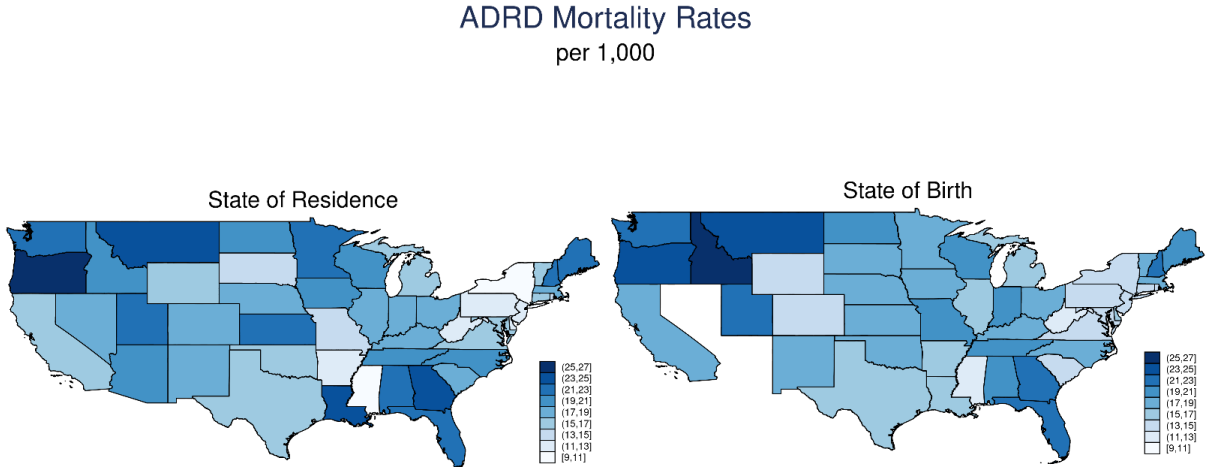
	(1) ADRD mortality	(2) ADRD mortality	(3) ADRD mortality	(4) ADRD mortality
Random effects	None	SOR	SOB	SOR & SOB
Female (*reference: male*)	1.509***	1.514***	1.508***	1.511***
	(0.04923)	(0.04945)	(0.04923)	(0.04933)
nH Black (*reference: nH White*)	0.6897***	0.6812***	0.6545***	0.6619***
	(0.04813)	(0.04888)	(0.04751)	(0.04729)
Hispanic (*reference: nH White*)	0.7314*	0.6964**	0.6958**	0.6843**
	(0.09129)	(0.08993)	(0.09111)	(0.08954)
Other race/ethnicity (*reference: nH White*)	0.7976	0.751	0.7401	0.748
	(0.1279)	(0.1313)	(0.1271)	(0.131)
Less than high school (*reference: college*)	0.7921***	0.7995***	0.7858***	0.7974***
	(0.04127)	(0.04197)	(0.04111)	(0.04187)
High school (*reference: college*)	0.9494	0.9532	0.9485	0.9551
	(0.05088)	(0.05122)	(0.05088)	(0.05134)
Some college (*reference: college*)	0.9174	0.9063	0.9063	0.9105
	(0.05925)	(0.05861)	(0.0586)	(0.05888)
Constant	0.01994***	0.02022***	0.02078***	0.02016***
	(0.00272)	(0.002826)	(0.002876)	(0.002767)
N	253,000	253,000	253,000	253,000
$\ln(\sigma_u^2)$		0.03968***	0.01982***	0.03485***
		(0.01063)	(0.006801)	(0.009743)
σ_u		0.1992	0.1408	0.1867
ρ		0.0119	0.006	0.0105
AIC	42200	42100	42160	42150
BIC	42930	42840	42900	42890

Source: Authors' calculations using NLMS data at a Federal Statistical Research Data Center under FSRDC Project Number 2338 (CBDRB-FY23-P2338-R10359).

Notes: Exponentiated coefficients with standard errors in parenthesis. Age at interview and NLMS cohort fixed effects are included in all regressions (coefficients not reported). In Column 4, the random effect accounts for the unique combinations of state of residence and state of birth. nH = non-Hispanic; RE = Random Effects; SOR = State of Residence; SOB = State of Birth; AIC = Akaike Information Criterion; BIC = Bayesian Information Criterion; ADRD = Alzheimer's Disease and Related Dementias. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$.

Figure

Figure 1. Spatial variation in ADRD mortality by state of residence and state of birth



Source: Authors' calculations using NLMS data at a Federal Statistical Research Data Center under FSRDC Project Number 2338 (CBDRB-FY23-P2338-R10359).
Notes: Arizona and Nevada excluded for state of birth illustration in accordance with Census Bureau rounding rules. ADRD = Alzheimer's Disease and Related Dementias.

Online supplementary file

Table S1. Robustness check: Controlling for age as a continuous measure

	(1) No RE	(2) SOR	(3) SOB	(4) SOR & SOB
(Age-60)	1.139***	1.139***	1.139***	1.139***
	(0.00569)	(0.005695)	(0.005694)	(0.005695)
$(Age-60)^2$	0.9970***	0.9970***	0.9970***	0.9970***
	(0.0001827)	(0.0001828)	(0.0001827)	(0.0001828)
Female (*reference: male*)	1.507***	1.513***	1.506***	1.509***
	(0.04917)	(0.04938)	(0.04916)	(0.04927)
nH Black (*reference: nH White*)	0.6891***	0.6807***	0.6538***	0.6612***
	(0.04808)	(0.04884)	(0.04745)	(0.04724)
Hispanic (*reference: nH White*)	0.7311*	0.6957**	0.6949**	0.6835**
	(0.09124)	(0.08983)	(0.091)	(0.08945)
Other race/ethnicity (*reference: nH White*)	0.7959	0.7487	0.738	0.7461
	(0.1277)	(0.1309)	(0.1268)	(0.1307)
Less than high school (*reference: college*)	0.7920***	0.7994***	0.7857***	0.7973***
	(0.04126)	(0.04196)	(0.04109)	(0.04186)
High school (*reference: college*)	0.9500	0.9538	0.9491	0.9558
	(0.0509)	(0.05124)	(0.0509)	(0.05136)
Some college (*reference: college*)	0.9166	0.9054	0.9054	0.9096
	(0.05918)	(0.05855)	(0.05853)	(0.05882)
Constant	0.007354***	0.007457***	0.007656***	0.007423***
	(0.0008639)	(0.0009038)	(0.0009164)	(0.0008783)
N	253,000	253,000	253,000	253,000
$\ln(\sigma_u^2)$		0.03994***	0.01997***	0.03512***
		(0.01069)	(0.006837)	(0.009787)
σ_u		0.1998	0.1413	0.1874
ρ		0.012	0.006	0.0106
AIC	42180	42080	42140	42120
BIC	42500	42410	42470	42450

Source: Authors' calculations using NLMS data at a Federal Statistical Research Data Center under FSRDC Project Number 2338 (CBDRB-FY24-P2338-R11203).

Notes: Exponentiated coefficients with standard errors in parenthesis. Age at interview and NLMS cohort fixed effects are included in all regressions (coefficients not reported). nH = non-Hispanic; RE = Random Effects; SOR = State of Residence; SOB = State of Birth; AIC = Akaike Information Criterion; BIC = Bayesian Information Criterion. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$.

Table S2. Robustness check: Excluding survivors from the sample

	(1) No RE	(2) SOR	(3) SOB	(4) SOR & SOB
Female (*reference: male*)	1.761***	1.770***	1.762***	1.765***
	(0.05782)	(0.05816)	(0.05787)	(0.05802)
nH Black (*reference: nH White*)	0.6859***	0.6798***	0.6535***	0.6597***
	(0.04797)	(0.04891)	(0.04751)	(0.04723)
Hispanic (*reference: nH White*)	0.7953	0.7561*	0.7520*	0.7396*
	(0.09956)	(0.09789)	(0.09888)	(0.09718)
Other race/ethnicity (*reference: nH White*)	0.8654	0.8024	0.7934	0.8002
	(0.1393)	(0.1401)	(0.1362)	(0.1400)
Less than high school (*reference: college*)	0.6959***	0.7035***	0.6906***	0.7009***
	(0.03621)	(0.03688)	(0.03607)	(0.03675)
High school (*reference: college*)	0.8748*	0.8795*	0.8735*	0.8803*
	(0.04714)	(0.04752)	(0.04712)	(0.04759)
Some college (*reference: college*)	0.8684*	0.8588*	0.8570*	0.8617*
	(0.05634)	(0.05581)	(0.05567)	(0.05599)
Constant	0.02105***	0.02136***	0.02193***	0.02125***
	(0.002874)	(0.00299)	(0.003038)	(0.002919)
N	180,000	180,000	180,000	180,000
$\ln(\sigma_u^2)$		0.04136***	0.02076***	0.03583***
		(0.01101)	(0.007052)	(0.009925)
σ_u		0.2034	0.1441	0.1893
ρ		0.0124	0.0063	0.0108
AIC	39970	39860	39930	39910
BIC	40670	40580	40640	40630

Source: Authors' calculations using NLMS data at a Federal Statistical Research Data Center under FSRDC Project Number 2338 (CBDRB-FY24-P2338-R11203).

Notes: Exponentiated coefficients with standard errors in parenthesis. Age at interview and NLMS cohort fixed effects are included in all regressions (coefficients not reported). nH = non-Hispanic; RE = Random Effects; SOR = State of Residence; SOB = State of Birth; AIC = Akaike Information Criterion; BIC = Bayesian Information Criterion. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$.